A CROSS-MODAL ATTENTION FRAMEWORK FOR INTERPRETABLE AGE-RELATED MACULAR DEGENERATION GRADING USING FUNDUS AND OCT IMAGE FUSION

A PROJECT REPORT

Submitted by

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in partial fulfillment of the requirements for the award degree of

Bachelor of Engineering

20CS8504 PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM – 621112

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BONAFIDE CERTIFICATE

The work embodied the project report titled **"A CROSS MODAL ATTENTION FRAMEWORK FOR INTERPRETABLE AGE-REALTED MASCULAR DEGENRATION GRADING USING FUNDUS AND OCT IMAGE FUSION"** has been carried out by the students **KALAIYARASI M, KAVIYA N, MONISHA V, NISHANTHI V.** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Age-related macular degeneration (AMD) is one of the leading causes of vision loss among older adults worldwide, affecting the macula, the central part of the retina responsible for clear and detailed vision, making early detection and proper grading essential for effective treatment and prevention of severe vision loss. Traditionally, ophthalmologists diagnose AMD by manually examining retinal images such as fundus photographs and Optical Coherence Tomography (OCT) scans; however, manual analysis can be time-consuming. A multimodal deep learning approach can be used to automatically detect and grade AMD using both fundus and OCT images, where Convolutional Neural Networks (CNNs) extract important features from each modality and a cross-modal attention mechanism combines these features to capture relationships between retinal surface patterns and structural details, and the fused information is then used to classify images into different AMD stages. The proposed system improves the accuracy and reliability of AMD detection compared to traditional single-modality approaches. It also provides a fast and user-friendly solution that can assist ophthalmologists in early diagnosis and clinical decision-making.

Keywords: Age-Related Macular Degeneration, Deep Learning, multi-eye disease image, Optical Coherence Tomography, Fundus Imaging, Retinal Image, Deep Learning.

ACKNOWLEDGEMENT

We thank our **Dr. N. Vasudevan**, Principal, for his valuable suggestions and support during the course of my research work.

We thank our **Mr. R. Rajavarman**, Head of the Department, Assistant Professor (Sr. Grade), Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We wish to record my deep sense of gratitude and profound thanks to my Guide **Mrs. R. Jasmine**, Assistant Professor, Computer Science and Engineering, for her keen interest, inspiring guidance, constant encouragement with my work during all stages, to bring this thesis into fruition.

We are extremely indebted to our project coordinator **Mrs. M. Mathumathi**, Assistant Professor, Computer Science and Engineering, for her valuable suggestions and support during the course of my research work.

We also thank the faculty and non-teaching staff members of the Department Computer Science and Engineering, K Ramakrishnan College of Technology, Samayapuram, for their valuable support throughout the course of my research work.

Finally, we thank our parents, friends and our well wishes for their kind support.

SIGNATURE

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LIST OF ABBREVIATIONS

AMD	-	Age-Related Macular Degeneration
OCT	-	Optical Coherence Tomography
CNN	-	Convolutional Neural Network
MLP	-	Multilayer Perceptron
AI	-	Artificial Intelligence
MSA-Net	-	Multi-Scale Attention Network
DSBN	-	Domain-Specific Batch Normalization
CABNet	-	Category Attention Block Network
EHR	-	Electronic Health Record

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

Age-Related Macular Degeneration (AMD) is a common eye disease that mainly affects older adults and can lead to permanent vision loss if not detected early. It damages the macula, the central part of the retina responsible for sharp and detailed vision. Since AMD progresses gradually, early detection and accurate grading are important for treatment and prevention of vision problems.

In medical practice, ophthalmologists use retinal imaging techniques such as fundus photography and Optical Coherence Tomography (OCT) to detect AMD. Fundus images show the surface of the retina and help identify abnormalities like drusen deposits, pigmentation changes, and bleeding, while OCT images provide cross-sectional views of the retinal layers to observe structural changes such as retinal thinning, fluid accumulation, and tissue damage. Although these techniques provide important information, manual analysis can be time-consuming.

To improve this, an automated system using deep learning can be employed, where both fundus and OCT images are used as input to capture different retinal features, followed by preprocessing techniques such as image resizing and normalization. The extracted features are then combined using a cross-modal attention mechanism, enabling the model to focus on important retinal regions and learn relationships between surface and structural features, and finally, the system predicts the stage of AMD and displays the result through a graphical user interface, assisting ophthalmologists in faster and more reliable diagnosis.

1.2 OBJECTIVE OF THE PROJECT

The main objective of this project is to develop an efficient and reliable system for the automatic detection and grading of Age-Related Macular Degeneration (AMD) using retinal images. AMD is a serious eye disease that can lead to vision loss if not detected at an early stage. Therefore, early diagnosis is very important for proper treatment and prevention of severe vision damage.

This project aims to design a deep learning-based model that can analyze retinal images such as Optical Coherence Tomography (OCT) images and fundus images. By using these images, the system can identify important features related to AMD and classify the condition of the retina. The use of both OCT and fundus images helps in capturing both surface-level and structural information of the retina, which improves the overall accuracy of detection.

Another important objective of this project is to reduce the dependency on manual diagnosis. Traditionally, ophthalmologists examine retinal images manually, which can be time-consuming and may vary depending on the experience of the doctor. The proposed system provides an automated solution that can quickly analyze images and produce accurate results, thereby saving time and supporting doctors in decision making.

The project also aims to build a user-friendly interface where users can easily upload retinal images and obtain prediction results. The system allows users to upload OCT images, fundus images, or both, making it flexible and easy to use. The final output includes the predicted result along with a confidence score, which helps in understanding the reliability of the prediction.

In addition, the system is designed to improve the accuracy of AMD detection by using advanced deep learning models such as Convolutional Neural Networks (CNN). These models are capable of learning complex patterns in medical images and provide better performance compared to traditional methods.

1.3 SCOPE OF THE PROJECT

The scope of this project focuses on developing an automated system for detecting and classifying Age-Related Macular Degeneration (AMD) using retinal images. The system is designed to assist in early diagnosis by analyzing medical images such as fundus photographs and Optical Coherence Tomography (OCT) scans. By using deep learning techniques, the system can identify patterns in retinal images and provide accurate predictions about the presence of AMD.

The project mainly covers the use of image processing and deep learning methods for medical image analysis. It includes preprocessing of retinal images, feature extraction using Convolutional Neural Networks (CNN), and classification of images into normal or AMD categories. The system also supports the use of both OCT and fundus images, which helps improve detection accuracy by combining different types of retinal information.

Another important scope of this project is the development of a user-friendly web-based interface. Users can upload retinal images through the system and receive prediction results quickly. This makes the system easy to use for both medical professionals and non-technical users. The system also displays results with confidence scores, which helps in understanding the reliability of the prediction.

The project can be extended for real-world applications in hospitals, clinics, and telemedicine platforms. It can assist ophthalmologists in screening large numbers of patients efficiently and help in early detection of AMD. This can reduce the workload of doctors and improve patient care.

In the future, the system can be enhanced by including more retinal diseases such as diabetic retinopathy and glaucoma. It can also be integrated with real-time medical imaging devices and electronic health record systems for better clinical support.

CHAPTER 2

LITERATURE SURVEY

2.1 AUTOMATIC DETECTION OF DIABETIC RETINOPATHY

This research presents a review of automatic detection techniques for Diabetic Retinopathy (DR) using retinal fundus images. Diabetic Retinopathy is a serious eye disease caused by damage to the blood vessels of the retina due to diabetes and can lead to severe vision loss if not detected early. Traditionally, ophthalmologists diagnose diabetic retinopathy by manually examining retinal fundus images to identify abnormalities such as microaneurysms, hemorrhages, and exudates; however, manual diagnosis can be time-consuming.

To address these challenges, automated detection approaches based on machine learning and morphological image processing are reviewed. Retinal images are preprocessed to enhance image quality and remove noise, followed by morphological operations to highlight retinal structures such as blood vessels and lesions, improving the visibility of disease-related features. After preprocessing, machine learning algorithms classify the retinal images into normal and diabetic retinopathy categories by learning patterns from features.

The study concludes that combining machine learning with morphological techniques improves detection performance and assists ophthalmologists in early diagnosis, reducing the risk of vision loss. Integrating such intelligent systems into clinical practice can reduce workload for ophthalmologists, provide faster screening, and support large-scale population-level eye disease monitoring programs.

2.2 HYPERPARAMETER TUNING DEEP LEARNING FOR DIABETIC RETINOPATHY FUNDUS IMAGE CLASSIFICATION

This research focuses on improving the classification of Diabetic Retinopathy (DR) using deep learning techniques applied to retinal fundus images. Diabetic Retinopathy is a serious eye disease caused by damage to the retinal blood vessels due to diabetes. Early detection of this disease is very important to prevent severe vision loss. Traditionally, ophthalmologists diagnose DR by manually examining fundus images to identify abnormalities such as microaneurysms, hemorrhages, and exudates.

In this study, the authors propose a deep learning approach that uses Multilayer Perceptron (MLP) models for classifying retinal fundus images. The system focuses on improving classification performance by applying hyperparameter tuning techniques. Hyperparameters are important parameters that control how a machine learning model learns from data. Proper tuning of these parameters helps improve the accuracy and efficiency of the classification model.

Initially, the retinal fundus images undergo preprocessing steps such as image enhancement and normalization to improve the quality of the images. Important features are extracted from the images, and these features are then used as input for the MLP model. The multilayer perceptron consists of multiple hidden layers that help the system learn complex patterns from the retinal images.The authors apply hyperparameter tuning methods to optimize the performance of the MLP model.

By adjusting parameters such as learning rate, number of hidden layers, and number of neurons, the model achieves better classification accuracy. The system is trained using a dataset of retinal fundus images representing different stages of diabetic retinopathy.The results of the study show that hyperparameter tuning significantly improves the performance of the deep learning model for fundus image classification.

2.3 CABNET: CATEGORY ATTENTION BLOCK FOR IMBALANCED DIABETIC RETINOPATHY GRADING

This research focuses on improving the grading of Diabetic Retinopathy (DR) using a deep learning model called Category Attention Block Network (CABNet). Diabetic Retinopathy is a serious eye disease that occurs due to damage to the blood vessels in the retina. Early detection and accurate grading of the disease are important to prevent severe vision loss. However, one of the major challenges in DR detection is the imbalanced dataset problem, where some disease categories have more training samples than others. This imbalance can affect the performance of deep learning models and reduce classification accuracy.

To address this problem, the authors propose a deep learning framework called CABNet, which incorporates a Category Attention Block mechanism. The proposed model focuses on improving feature representation by giving more importance to important disease-related features in retinal images. The attention mechanism helps the model focus on specific regions of the retina that contain abnormalities such as microaneurysms, hemorrhages, and exudates.

In the proposed system, retinal fundus images are first preprocessed to enhance image quality and remove noise. The images are then passed through the CABNet model, which extracts important features from different layers of the neural network. The Category Attention Block helps balance the learning process by focusing on underrepresented disease categories, improving the overall performance of the model.The attention mechanism allows the system to learn more meaningful features from retinal images, which helps in better classification of different disease stages. This research demonstrates that attention-based deep learning models can effectively address the problem of class imbalance in system.

2.4 DIABETIC RETINOPATHY DIAGNOSIS FROM FUNDUS IMAGES USING STACKED GENERALIZATION OF DEEP

This research focuses on the automatic diagnosis of Diabetic Retinopathy (DR) using retinal fundus images and deep learning techniques. Diabetic Retinopathy is a common complication of diabetes that affects the blood vessels in the retina and can lead to severe vision loss if not detected early. Traditionally, ophthalmologists diagnose DR by manually analysing retinal images to identify abnormalities such as microaneurysms, hemorrhages, and exudates. However, manual diagnosis is time-consuming and depends on the expertise of medical professionals.

In this study, the authors propose a deep learning approach based on Convolutional Neural Networks (CNNs) combined with a stacked generalization technique to improve classification accuracy. Stacked generalization is an ensemble learning method that combines multiple deep learning models to produce better predictive performance. Instead of relying on a single model, the system integrates the outputs of several CNN models to make a final prediction.

Initially, retinal fundus images are preprocessed to enhance image quality and remove noise. The images are then passed through multiple CNN models that extract important features related to retinal abnormalities. Each CNN model learns different patterns from the dataset, and their predictions are combined using the stacked generalization approach. The experimental results show that the proposed stacked deep learning model improves the accuracy of diabetic retinopathy detection compared to single CNN models. The study demonstrates that ensemble deep learning techniques can effectively enhance retinal disease detection systems.

2.5 A BENCHMARK FOR STUDYING DIABETIC RETINO: SEGMENTATION, GRADING, AND TRANSFERABILITY

This research focuses on developing a benchmark framework for studying Diabetic Retinopathy (DR) using retinal fundus images. Diabetic Retinopathy is a major eye disease that occurs due to damage to the retinal blood vessels caused by diabetes. Early detection and accurate grading of the disease are essential to prevent severe vision loss. However, developing automated detection systems for DR is challenging because retinal image datasets often vary in quality, size, and distribution.

In this study, the authors propose a benchmark framework that evaluates different tasks related to diabetic retinopathy, including segmentation, disease grading, and model transferability. Segmentation involves identifying and isolating important regions in retinal images, such as blood vessels and lesion areas. Disease grading refers to classifying the severity of diabetic retinopathy based on retinal abnormalities.

The proposed system uses a technique called Domain-Specific Batch Normalization (DSBN) to improve the performance of deep learning models when working with multiple datasets. DSBN allows the neural network to adapt to different data distributions by applying separate normalization parameters for each dataset domain. This helps the model learn more robust features and improves its ability to generalize across different retinal image datasets.

The authors evaluate the performance of the proposed framework using multiple retinal image datasets and compare the results with existing methods. The experimental results show that the use of Domain-Specific Batch Normalization improves the accuracy and transferability of deep learning models for diabetic retinopathy detection and grading.

2.6 AUTOMATED BINARY AND MULTICLASS OF DIABETIC RETINOPATHY USING HARALICK AND MULTIRESOLUTION FEATURES

This research focuses on the automatic detection and classification of Diabetic Retinopathy (DR) using retinal fundus images and advanced feature extraction techniques. Diabetic Retinopathy is a serious eye disease caused by damage to the blood vessels in the retina due to diabetes. Early detection of the disease is very important to prevent vision loss. Traditionally, ophthalmologists diagnose DR by manually analyzing retinal images to identify abnormalities such as microaneurysms, hemorrhages, and exudates. However, manual diagnosis can be time-consuming and may lead to errors due to human limitations.In this study, the authors propose an automated system that performs binary and multiclass classification of diabetic retinopathy using texture and multiresolution features extracted from retinal images

In addition, the study uses Anisotropic Dual-Tree Complex Wavelet Transform (ADTCWT) to extract multiresolution features from the retinal images. This technique helps analyze the image at different scales and orientations, allowing the system to capture both fine and coarse details of retinal abnormalities. By combining Haralick texture features with multiresolution features obtained through ADTCWT, the system can extract more meaningful information from the retinal images.

The extracted features are then used to classify retinal images into different categories such as normal and different stages of diabetic retinopathy. The experimental results show that the proposed approach improves the accuracy of both binary and multiclass classification tasks. The study demonstrates that combining texture features with multiresolution analysis techniques can significantly enhance the performance of automated diabetic retinopathy detection systems.

2.7 MULTI-SCALE ATTENTION NETWORK FOR DIABETIC RETINOPATHY CLASSIFICATION

This research focuses on the automatic classification of Diabetic Retinopathy (DR) using retinal fundus images and a deep learning-based attention model. Diabetic Retinopathy is a major eye disease caused by damage to retinal blood vessels due to diabetes. If the disease is not detected early, it can lead to severe vision loss. Traditionally, ophthalmologists diagnose DR by manually examining retinal fundus images to identify abnormalities such as microaneurysms, hemorrhages, and exudates. However, manual analysis is time-consuming and depends heavily on the expertise of medical professionals.

To overcome these limitations, the authors propose a deep learning framework called Multi-Scale Attention Network (MSA-Net) for diabetic retinopathy classification. The proposed model is designed to capture retinal features at multiple scales, which helps detect both small and large abnormalities present in retinal images. By analyzing features at different scales, the system can identify disease-related patterns more effectively.

In the proposed approach, retinal fundus images are first preprocessed to enhance image quality and remove noise. The images are then passed through the multi-scale attention network, which extracts important features from different layers of the neural network. The attention mechanism allows the model to focus on the most relevant regions of the retinal image, such as areas containing lesions or abnormal blood vessels.

The multi-scale feature extraction process enables the system to analyze retinal abnormalities of different sizes, improving the accuracy of diabetic retinopathy classification. The experimental results show that the MSA-Net model achieves better performance compared to traditional deep learning models for retinal disease detection.

2.8 MULTIMODAL DEEP LEARNING FOR AUTOMATED AGE-RELATED MACULAR DEGENERATION CLASS USING FUNDUS AND OCT IMAGES

This research focuses on the automatic classification of Age-Related Macular Degeneration (AMD) using retinal images obtained from both fundus photography and Optical Coherence Tomography (OCT). AMD is a progressive retinal disease that affects the macula and is one of the leading causes of vision loss among older adults. Early detection and accurate classification of AMD are important for proper treatment and disease management. Traditionally, ophthalmologists analyze fundus and OCT images manually to identify abnormalities such as drusen deposits, retinal layer damage, and fluid accumulation. However, manual analysis can be time-consuming and may depend on the expertise of the medical professional.

In this study, the authors propose a multimodal deep learning framework that combines information from fundus and OCT images to improve AMD classification accuracy. The proposed system uses deep learning models to extract features from both imaging modalities separately. Fundus images provide information about the surface structure of the retina, while OCT images provide cross-sectional views of the retinal layers. By analyzing both image types together, the system can capture more comprehensive retinal information.

The proposed method uses feature-level fusion to combine the extracted features from both modalities. In this approach, important features from fundus images and OCT images are integrated into a single feature representation before classification. This multimodal feature fusion allows the model to learn the relationships between surface retinal features and internal retinal structures. As a result, the system can identify disease patterns more effectively than single-modality models.

2.9 ATTENTION-BASED MULTIMODAL NETWORK FOR AGE-RELATED MACULAR DEGENERATION DIAGNOSIS USING FUNDUS PHOTOGRAPHY AND OCT

This research focuses on improving the diagnosis of Age-Related Macular Degeneration (AMD) using a multimodal deep learning approach that combines fundus photography and Optical Coherence Tomography (OCT) images. AMD is a common eye disease that affects the macula and can lead to severe vision loss in older adults. Early detection and accurate diagnosis are essential for effective treatment and disease management. Traditionally, ophthalmologists analyze fundus and OCT images separately to detect abnormalities such as drusen deposits, retinal layer damage, and fluid accumulation.

To address these challenges, the authors propose an attention-based multimodal Convolutional Neural Network (CNN) for automated AMD diagnosis. The proposed system extracts features separately from fundus and OCT images using deep learning models. Fundus images provide information about the surface structure of the retina, while OCT images reveal the internal structure of retinal layers. By combining information from both imaging modalities, the system can capture more detailed retinal features associated with AMD.

The key component of the proposed model is the attention mechanism, which helps the network focus on important regions in the retinal images. The attention module assigns higher importance to relevant features such as drusen, lesions, and structural abnormalities. This allows the system to improve feature representation and enhance the accuracy of disease detection.

The experimental results show that the attention-based multimodal network significantly improves AMD diagnosis accuracy compared to single-modality models.

2.10 INTERPRETABLE MULTIMODAL LEARNING FOR AGE RELATED MACULAR DEGENERATION DETECTION USING FUNDUS AND OCT IMAGES

This research focuses on developing an interpretable multimodal deep learning framework for detecting Age-Related Macular Degeneration (AMD) using both fundus images and Optical Coherence Tomography (OCT) scans. AMD is a progressive retinal disease that affects the macula and is one of the major causes of vision loss among elderly people. Early detection of AMD is very important for effective treatment and prevention of severe vision damage. Traditionally, ophthalmologists diagnose AMD by manually analyzing retinal images, which requires significant experience and can be time-consuming.

In this study, the authors propose a cross-modal attention-based deep learning model that combines information from fundus and OCT images to improve AMD detection. Fundus images provide information about the surface structure of the retina, including drusen deposits and pigmentation changes, while OCT images provide cross-sectional views of the retinal layers and reveal structural abnormalities. By analysing both imaging modalities together, the system can capture more comprehensive retinal information.

The proposed system uses a cross-modal attention network to integrate features extracted from both fundus and OCT images. The attention mechanism helps the model learn relationships between the two imaging modalities and focus on the most important regions related to AMD. This improves the model's ability to detect disease patterns and increases the accuracy of classification.

Another important feature of the proposed approach is explainability. The system provides visual explanations that highlight the regions of retinal images that contribute most to the model's predictions. This helps ophthalmologists understand how the model makes its decisions and increases trust in the automated diagnostic system.

CHAPTER 3

EXISTING SYSTEM

In recent years, several automated systems have been developed for detecting retinal diseases such as Age-Related Macular Degeneration (AMD) using medical imaging technologies. These systems mainly rely on retinal images such as fundus photographs or Optical Coherence Tomography (OCT) scans to analyze abnormalities in the retina. Machine learning and deep learning techniques are widely used to process these images and identify disease-related patterns. Although these systems have improved the efficiency of disease detection, many existing approaches still have certain limitations.

Most existing AMD detection systems are designed to work with only one imaging modality, either fundus images or OCT images. Fundus image-based systems analyze visible abnormalities on the retinal surface, such as drusen deposits, pigmentation changes, and blood vessel irregularities. On the other hand, OCT-based systems focus on analyzing cross-sectional retinal layers to detect structural changes such as retinal thinning, fluid accumulation, and tissue damage. Since each imaging modality provides different types of information, relying on a single modality may result in incomplete analysis of retinal conditions.

Another limitation of existing systems is that fundus and OCT images are often processed independently. Many deep learning models analyze these images separately and do not effectively combine the information from both modalities. As a result, the system may fail to capture the relationship between surface retinal features and internal structural changes. This can reduce the accuracy of disease detection and classification.

Furthermore, some traditional machine learning approaches require manual feature extraction, where experts must identify important image features before training the model. This process can be time-consuming and may not capture all relevant disease patterns. Even though deep learning models automatically extract features, single-

modality models may still miss important information that appears in other imaging modalities.

Because of these limitations, existing systems may not provide the most accurate or reliable diagnosis of AMD. Therefore, there is a need for advanced approaches that can integrate information from multiple retinal imaging modalities and improve the performance of automated disease detection systems.

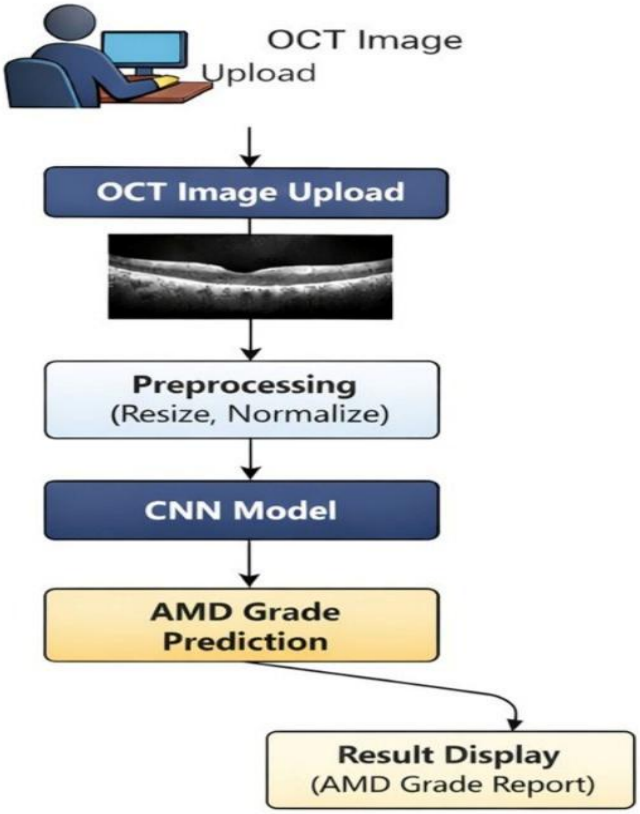

Figure 3.1 EXISTING SYSTEM DIAGRAM

CHAPTER 4

PROBLEMS IDENTIFIED

Existing systems for detecting Age-Related Macular Degeneration (AMD) using retinal images face several challenges related to accuracy, feature extraction, and efficient use of multiple imaging modalities. Although many machine learning and deep learning models have been developed for retinal disease detection, most of these systems still have limitations that affect their reliability and performance in clinical diagnosis.

One of the major problems in existing AMD detection systems is the use of a single imaging modality. Many automated systems rely only on fundus images or only on OCT images for disease detection. Fundus images provide information about the surface structure of the retina, while OCT images provide cross-sectional views of the retinal layers. When only one imaging modality is used, the system cannot capture complete information about retinal abnormalities. As a result, important disease-related features may be missed, which can reduce the accuracy of AMD detection.

Another significant issue is the separate analysis of fundus and OCT images. Even when both types of retinal images are available, many existing systems process them independently without integrating the information from both modalities. Since fundus images reveal surface-level features and OCT images reveal internal retinal structures, analyzing them separately limits the system's ability to understand the relationship between these features. This lack of multimodal integration reduces the overall effectiveness of disease detection systems.

Existing systems also face difficulties in feature extraction and model generalization. Some traditional machine learning models require manual feature extraction, where experts must identify important retinal features before training the model. This process can be time-consuming and may not capture all relevant patterns in retinal images.

CHAPTER 5

PROPOSED SYSTEM

The proposed system introduces a multimodal deep learning framework for automatic detection and grading of Age-Related Macular Degeneration (AMD) using retinal images. The system combines information from fundus images and Optical Coherence Tomography (OCT) images to improve the accuracy and reliability of AMD diagnosis. Unlike traditional systems that rely on a single imaging modality, the proposed approach integrates both surface-level and structural retinal information to provide a more comprehensive analysis of retinal abnormalities.

After the images are uploaded, the image preprocessing module prepares the images for analysis. In this stage, several preprocessing techniques are applied, including image resizing, normalization, and enhancement. All images are resized to a fixed resolution to ensure consistency during model training and testing. Image normalization is applied to adjust pixel intensity values, which helps improve the performance of the deep learning model.

Following preprocessing, the images are passed to the feature extraction module, where deep learning models extract important features from retinal images. In this system, Convolutional Neural Networks (CNNs) are used to automatically learn hierarchical features from both fundus and OCT images. Two parallel CNN branches are used, where one branch processes fundus images to extract surface retinal features such as drusen and pigmentation changes, while the other branch processes OCT images to extract structural features such as retinal layer thickness and fluid accumulation.

The next stage of the system is multimodal feature fusion, where the extracted features from both imaging modalities are combined. This attention mechanism helps the model focus on the most relevant retinal regions and learn the relationship between features extracted from fundus and OCT images. By combining complementary information from both modalities, the system can capture more detailed disease patterns.

Once the features are fused, they are passed to the classification module, where the system predicts the stage of AMD. The classification model analyzes the combined features and categorizes the retinal images into different classes such as Normal, Early AMD, Intermediate AMD, or Advanced AMD. This automated classification helps doctors quickly understand the severity of the disease.

Overall, the proposed system provides several advantages compared to existing approaches. By integrating multimodal retinal images and using advanced deep learning techniques, the system improves the accuracy of AMD detection and grading. This automated diagnostic tool can assist ophthalmologists in early disease detection, reduce diagnostic workload, and support more reliable clinical decision-making.

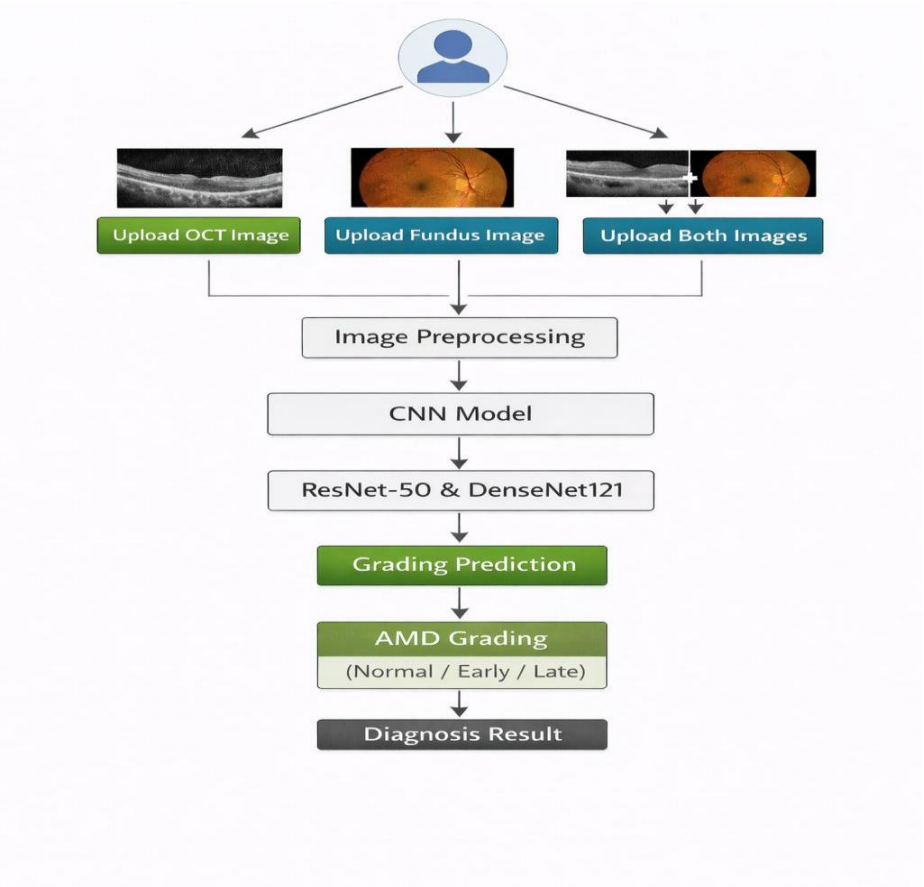

Figure 5.1 PROPOSED SYSTEM DIAGRAM

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

Processor	Intel Core i5 or higher
RAM	8 GB
Hard disk	256 GB or higher
System Type	64-bit Computer

6.2 SOFTWARE REQUIREMENTS

Operating System	Windows 11
Programming languages	Python 3.11
Frontend	HTML, CSS, Bootstrap
Backend	Python (Flask)
Database	SQLite
Libraries	Torchvision , Pillow, NumPy, Werkzeug
IDE/Tool	VS Code

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

- Image Upload Module
- Image Preprocessing Module
- Feature Extraction Module
- Muti-modal Feature Fusion Module
- AMD Classification Module

7.2 MODULE DESCRIPTION

7.2 .1 Image Upload Module

The Image Upload Module is an important component of the AMD detection system as it allows users to upload retinal images for analysis. In this module, users can upload fundus images, OCT images, or both types of images through the system interface. The uploaded images serve as the input data for the deep learning model that will analyse the retinal condition.

The module ensures that the uploaded images are in supported formats such as JPEG or PNG and verifies the file size and image quality before processing. This validation step is important to ensure that only valid and usable retinal images are accepted by the system. Once the images are successfully uploaded, they are securely stored in the system for further processing.

The image upload module also helps maintain an organized dataset by storing images in appropriate directories or databases. This allows the system to easily retrieve and process the images during the analysis stage. By providing a simple and user-friendly interface for uploading retinal images, this module

ensures that users can easily interact with the system and submit the required data for AMD detection.

7.2.2 Image Preprocessing Module

The Image Preprocessing Module prepares the uploaded retinal images for deep learning analysis. Medical images often contain noise, variations in lighting, and differences in resolution that can affect the performance of machine learning models. Therefore, preprocessing techniques are applied to improve image quality and ensure consistency across the dataset.

In this module, all retinal images are resized to a fixed resolution, typically 224 × 224 pixels, to match the input requirements of the convolutional neural network. Image normalization is also applied to adjust the pixel intensity values and improve the contrast of the images.

In addition, data augmentation techniques such as image rotation, horizontal flipping, and brightness adjustment may be applied to increase the diversity of the training dataset. These techniques help the model learn more robust features and reduce the risk of overfitting. By improving image quality and standardizing the input data, the preprocessing module plays a crucial role in enhancing the performance of the deep learning model.

7.2.3 Feature Extraction Module

The Feature Extraction Module is responsible for identifying important patterns and characteristics from retinal images. In this system, Convolutional Neural Networks (CNNs) are used to automatically extract meaningful features from both fundus and OCT images.

Two parallel CNN branches are used in the system. One branch processes fundus images, which capture the surface structure of the retina, including blood vessels, pigmentation changes, and drusen deposits. The other branch processes

OCT images, which provide cross-sectional views of retinal layers and help identify structural abnormalities such as retinal thinning and fluid accumulation.

The CNN model consists of multiple layers such as convolution layers, pooling layers, and activation functions that work together to learn hierarchical image features. The early layers detect simple features such as edges and textures, while deeper layers capture more complex patterns related to retinal diseases. The extracted features are then passed to the next stage for multimodal fusion.

7.2.4 Multi-modal Feature Fusion Module

The Multimodal Feature Fusion Module combines the features extracted from fundus and OCT images to create a more comprehensive representation of retinal abnormalities. Since each imaging modality provides different types of information, combining them allows the system to analyze both surface retinal features and internal retinal structures simultaneously.

In this module, a cross-modal attention mechanism is used to integrate the features obtained from the two CNN branches. The attention mechanism helps the model focus on the most relevant features from each modality and learn the relationship between them.

By assigning different importance weights to different features, the attention mechanism improves the quality of the fused feature representation. This helps the system capture more detailed disease patterns and enhances the overall performance of AMD detection and classification.

7.2.5 AMD Classification Module

The AMD Classification Module is a critical component of the proposed system, as it is responsible for determining the stage of Age-Related Macular Degeneration (AMD) based on the features extracted and fused from fundus and OCT images. After the multimodal feature fusion process is completed, the combined feature representation is forwarded to the classification stage where the

system analyzes the patterns associated with retinal abnormalities. This module uses deep learning-based classification techniques to accurately determine the severity of the disease.

In this stage, the fused feature vectors obtained from the previous module are passed through a series of fully connected neural network layers. These layers learn complex relationships between the extracted features and different stages of AMD. The classification model processes the features and identifies patterns related to retinal changes such as drusen deposits, retinal layer damage, and structural abnormalities visible in fundus and OCT images.

The final layer of the classification network uses a softmax activation function, which calculates the probability values for each possible disease category. Based on these probabilities, the system predicts the most likely AMD stage. The classification results are typically categorized into multiple classes such as Normal, Early AMD, Intermediate AMD, and Advanced AMD. This multi-class classification approach allows the system to provide more detailed diagnostic information rather than simply identifying whether the disease is present or not.

Another important aspect of this module is its ability to learn from large datasets of retinal images during the training phase. By analyzing numerous examples of normal and diseased retinal images, the deep learning model becomes capable of recognizing subtle patterns that may indicate the presence or progression of AMD. This improves the accuracy and reliability of the classification process.

CHAPTER 8

SYSTEM TESTING

8.1 UNIT TESTING

Unit Testing is performed to verify that each individual component or module of the AMD detection system works correctly and performs its intended function. In the proposed system, different modules such as the image upload module, image preprocessing module, feature extraction module, multimodal feature fusion module, AMD classification module, and result visualization module are tested independently. The purpose of this testing is to ensure that each module produces the expected output when given valid input.

For example, the image upload module is tested to confirm that retinal images such as fundus and OCT images can be uploaded successfully and stored in the system. The image preprocessing module is tested to verify whether the system correctly resizes and normalizes the images before sending them to the deep learning model. Similarly, the feature extraction module is tested to ensure that the convolutional neural network correctly extracts meaningful features from the retinal images.

The AMD classification module is tested to verify that the system correctly predicts the class of the retinal image based on the extracted features. The model is checked using different input images to ensure that it produces accurate and consistent results such as normal or AMD. The output probabilities and confidence scores are also verified to confirm that the classifier is functioning correctly. This helps ensure that the model can reliably identify retinal conditions.

The result visualization module is tested to ensure that the final prediction is displayed clearly and correctly to the user through the system interface. The system is checked to confirm that the predicted result, along with the confidence score, is shown in a user-friendly format.

8.2 INTEGRATION TESTING

Integration Testing is conducted to verify that different modules of the system work together correctly after they are combined. In the online voting application, multiple modules such as user registration, login authentication, candidate listing, vote casting, election management, and blockchain storage must interact with each other seamlessly. During integration testing, the connections between these modules are examined to ensure that data flows properly from one component to another without errors.

For example, after a user registers successfully, the system should allow the user to log in using the registered credentials. Similarly, once a voter logs in and casts a vote, the vote should be securely transmitted to the blockchain module and stored permanently in the blockchain database. Integration testing also verifies that the frontend interface communicates properly with the backend server and that APIs are functioning correctly. Any issues related to data transfer, communication errors, or incorrect system responses are identified and corrected during this phase. This testing ensures that the complete workflow of the voting system operates smoothly and that all modules interact correctly to provide a reliable and efficient voting experience.

In the AMD detection system, integration testing ensures that all modules such as image upload, preprocessing, feature extraction, classification, and result visualization work together smoothly. For example, after the user uploads fundus or OCT images, the system should automatically pass the images to the preprocessing module. The processed images are then forwarded to the feature extraction module, where the deep learning model analyzes them and extracts important features. These features are then sent to the classification module to predict whether the image is normal or affected by AMD.

Integration testing also verifies that the frontend interface correctly communicates with the backend system. When the user clicks the predict button, the request should be properly sent to the server, and the prediction result should be returned without any errors.

8.3 SYSTEM TESTING

System Testing is performed to evaluate the complete and fully integrated system to ensure that it meets the specified requirements and functions correctly in a real-world environment. In the online voting system, system testing involves verifying the entire workflow of the application, starting from user registration and login to vote casting and result management. The goal of this testing is to ensure that all components of the system work together as a complete system without any errors or performance issues. During system testing, various scenarios are tested such as registering a new user, authenticating the user through biometric verification, displaying the candidate list, casting a vote, and storing the vote securely in the blockchain. The system is also tested to confirm that each voter can vote only once and that unauthorized users cannot access the voting system. In addition, the administrator functions such as election creation, candidate management, and monitoring of election results are tested to ensure that they operate correctly. System testing helps verify the overall performance, reliability, security, and functionality of the voting application before it is deployed for actual use.

In the AMD detection system, system testing is carried out to verify the complete workflow of the application, starting from image upload to final result display. Different test cases are used to check whether the system correctly processes fundus and OCT images and produces accurate predictions. The system is tested to ensure that images are uploaded successfully, preprocessing is applied correctly, and the deep learning model analyzes the images without errors. This helps confirm that the system functions properly in real-time conditions.

The system is also tested for performance and usability to ensure that users can easily interact with the application. The response time of the system is evaluated to check how quickly the model generates predictions after image upload. Additionally, the user interface is tested to ensure that results such as AMD detection and confidence score are displayed clearly. System testing ensures that the application is reliable, efficient, and ready for practical use in medical diagnosis.

8.4 SECURITY TESTING

Security Testing is one of the most important testing processes in an online voting system because the system handles highly sensitive data such as voter identities and voting results. The purpose of security testing is to identify vulnerabilities and ensure that the system is protected from unauthorized access, hacking attempts, and data manipulation. In this project, security testing verifies whether the user authentication process is secure and whether only authorized users can access the system. The system is tested to ensure that user data is encrypted and that votes stored in the blockchain cannot be modified or deleted by unauthorized users. Security testing also examines whether the application is protected against common cyber threats such as SQL injection, cross-site scripting, and unauthorized API access. Additionally, the blockchain technology used in the system is evaluated to ensure that the vote records are immutable and tamper-proof. By performing security testing, developers can identify weaknesses in the system and implement necessary protection mechanisms to ensure the safety, confidentiality, and integrity of the voting data.

In the AMD detection system, security testing is performed to ensure that all user data and uploaded retinal images are protected from unauthorized access. The system is tested to verify that only authenticated users can access the application and upload images for analysis. Proper validation checks are implemented to ensure that only valid image files are accepted, preventing the upload of harmful or unsupported files. This helps protect the system from potential security threats and ensures safe handling of medical data.

Security testing also ensures that communication between the frontend and backend is secure and that sensitive information is not exposed during data transmission. The system is checked for common vulnerabilities such as unauthorized access, data leakage, and improper input handling. Additionally, access control mechanisms are tested to ensure that only authorized users can view results or perform specific actions.

8.5 USER ACCEPTANCE TESTING

User Acceptance Testing is the final stage of testing in which the system is evaluated by real users to determine whether it meets their expectations and requirements. In the online voting system, this testing is conducted by allowing a group of users to interact with the system and perform tasks such as registering, logging in, viewing candidate details, and casting their votes. The purpose of user acceptance testing is to ensure that the system is user-friendly, reliable, and easy to understand for voters and administrators. During this testing phase, users provide feedback regarding the system interface, navigation, performance, and overall usability. Any issues related to system functionality, confusion in the voting process, or difficulties in accessing certain features are identified and corrected before the system is officially deployed. This testing ensures that the online voting application meets the needs of its intended users and provides a smooth and efficient voting experience.

In the AMD detection system, user acceptance testing is carried out by allowing users such as students, researchers, or medical professionals to interact with the application. Users are asked to upload fundus and OCT images, select prediction options, and view the results generated by the system. This helps evaluate whether the system is easy to use and whether the prediction results are clearly displayed and understandable.

During this phase, feedback is collected from users regarding the system interface, ease of navigation, speed of prediction, and clarity of results such as AMD detection and confidence score. Any issues such as difficulty in uploading images, unclear results, or slow response time are identified and improved. User acceptance testing ensures that the system is user-friendly, reliable, and suitable for real-world usage in retinal disease detection.

CHAPTER 9

RESULT AND DISCUSSION

The development and implementation of the Multimodal Deep Learning System for Age-Related Macular Degeneration (AMD) Detection using Fundus and OCT Images successfully demonstrates how artificial intelligence can assist in early detection and grading of retinal diseases. The system was designed with several modules such as image upload, image preprocessing, feature extraction, multimodal feature fusion, AMD classification, and result visualization to ensure accurate analysis of retinal images. After integrating all the modules and testing the system functionality, the results showed that the proposed system effectively improves the detection accuracy of AMD compared to traditional single-modality approaches.

The image upload module was successfully implemented to allow users to upload retinal images such as fundus photographs and OCT scans. The system was able to validate image formats and store them securely for further processing. During testing, the module correctly accepted valid retinal images and rejected unsupported file types. This ensured that only appropriate input data was used for analysis, improving the reliability of the system.

The image preprocessing module played an important role in preparing the retinal images for deep learning analysis. Techniques such as image resizing, normalization, and data augmentation were applied to improve image quality and maintain consistency in the dataset. During testing, the preprocessing module successfully converted all input images to the required size and format. This step helped improve the performance of the deep learning model by reducing noise and enhancing important retinal features.

The feature extraction module used Convolutional Neural Networks (CNNs) to automatically learn important features from fundus and OCT images. Two parallel CNN branches were implemented, where one branch processed fundus images and the other processed OCT images. The system successfully extracted relevant features related to

retinal abnormalities such as drusen deposits, pigmentation changes, and structural variations in retinal layers. The testing results showed that the deep learning model could effectively capture meaningful patterns from retinal images.

The multimodal feature fusion module integrated the features extracted from both imaging modalities using a cross-modal attention mechanism. This mechanism helped the model focus on the most important features from both fundus and OCT images. During testing, the system demonstrated improved feature representation by combining surface retinal features with internal structural information. This integration helped improve the overall accuracy of AMD detection compared to models that rely on a single imaging modality.

The AMD classification module successfully predicted the stage of the disease based on the fused features obtained from the previous modules. The classification model categorized the retinal images into different classes such as Normal, Early AMD, Intermediate AMD, and Advanced AMD. The system achieved reliable classification performance by analyzing patterns learned during the training process. This automated classification process helps doctors quickly identify the severity of the disease and supports early diagnosis.

The result visualization module displayed the predicted results clearly through the system interface. The system presented the detected AMD stage in a user-friendly format, allowing doctors and users to easily understand the diagnostic outcome. In some cases, the system also highlighted important regions of the retinal images that influenced the prediction, which helped improve the interpretability of the deep learning model.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

The development of the Multimodal Deep Learning System for Age-Related Macular Degeneration (AMD) Detection using Fundus and OCT Images provides an advanced approach for automated retinal disease diagnosis. Age-Related Macular Degeneration is one of the leading causes of vision loss among older adults, and early detection is essential to prevent severe vision impairment. Traditional diagnostic methods rely on manual examination of retinal images by ophthalmologists, which can be time-consuming and dependent on the experience of medical professionals. The proposed system addresses these challenges by applying artificial intelligence and deep learning techniques to automatically analyse retinal images and detect AMD.

The system was designed with several important modules including image upload, image preprocessing, feature extraction, multimodal feature fusion, AMD classification, and result visualization. Each module plays a significant role in ensuring that the system operates effectively and produces reliable diagnostic results. The image upload module allows users to provide retinal images such as fundus photographs and OCT scans for analysis. The preprocessing module enhances the quality of the images and prepares them for deep learning processing by applying resizing, normalization, and augmentation techniques.

The feature extraction module uses Convolutional Neural Networks (CNNs) to automatically learn important features from retinal images. Separate CNN branches process fundus images and OCT images, capturing both surface retinal features and structural information from retinal layers. The multimodal feature fusion module integrates these features using a cross-modal attention mechanism, which allows the model to focus on the most relevant regions of the retina. This integration improves the model's ability to detect complex patterns associated with AMD.

The classification module analyses the fused features and categorizes the retinal images into different disease stages such as Normal, Early AMD, Intermediate AMD, and Advanced AMD. The result visualization module displays the predicted results in a clear and understandable format, helping doctors interpret the diagnosis quickly. In some cases, the system can also highlight important retinal regions that influenced the prediction, improving the interpretability of the model.

The results obtained from system testing demonstrate that the proposed multimodal deep learning system improves the accuracy and reliability of AMD detection compared to traditional single-modality methods. By combining information from both fundus and OCT images, the system provides a more comprehensive analysis of retinal abnormalities.

In conclusion, the proposed system demonstrates that artificial intelligence and multimodal deep learning techniques can significantly improve automated retinal disease diagnosis. The system can assist ophthalmologists in early detection of AMD, reduce diagnostic workload, and support better clinical decision-making. Although further improvements can enhance the system's capabilities, the current implementation successfully demonstrates the potential of deep learning technologies in medical image analysis.

10.2 FUTURE ENHANCEMENTS

The proposed multimodal deep learning system for AMD detection provides a reliable platform for automated retinal disease diagnosis. However, several enhancements can be implemented in the future to further improve the system's performance, usability, and clinical applicability.

Future versions of the system can be trained using larger and more diverse retinal image datasets. A larger dataset would allow the deep learning model to learn more robust features and improve its ability to detect AMD in different populations and imaging conditions.

Currently, the system uses fundus and OCT images for AMD detection. In future research, additional imaging techniques such as fluorescein angiography or fundus autofluorescence can be integrated into the system. Combining multiple imaging modalities may provide more detailed retinal information and further improve diagnostic accuracy.

The system can be developed into a real-time clinical decision support tool that can be used in hospitals and ophthalmology clinics. By integrating the system with medical imaging devices, doctors could receive immediate diagnostic results while examining patients.

Future versions of the system can be deployed as mobile or web-based applications to increase accessibility. This would allow doctors and healthcare professionals to analyse retinal images remotely, enabling telemedicine services for patients in remote or underserved areas.

Deep learning models are often considered black-box systems. Future research can focus on improving model interpretability by integrating explainable AI techniques that provide detailed visual explanations of the model's predictions. This would help doctors better understand the decision-making process of the system.

APPENDIX – A

SOURCE CODE

dataset_util.py

```
import json

import random

from collections import defaultdict

from pathlib import Path

from typing import Dict, List, Tuple

import torch

from torch.utils.data import DataLoader, Subset, WeightedRandomSampler

from torchvision import datasets, transforms

def save_class_indices(class_indices: dict, output_path: Path) -> None:

    output_path.parent.mkdir(parents=True, exist_ok=True)

    with open(output_path, "w", encoding="utf-8") as f:

        json.dump(class_indices, f, indent=2)

def _stratified_split(

    targets: List[int],

    validation_split: float,

    seed: int,

) -> Tuple[List[int], List[int]]:

    by_class: Dict[int, List[int]] = defaultdict(list)

    for idx, label in enumerate(targets):
```

```
    by_class[label].append(idx)
rng = random.Random(seed)
train_indices: List[int] = []
val_indices: List[int] = []
for indices in by_class.values():
    rng.shuffle(indices)
    if len(indices) <= 1:
        train_indices.extend(indices)
        continue
    val_count = int(round(len(indices) * validation_split))
    val_count = max(1, min(val_count, len(indices) - 1))
    val_indices.extend(indices[:val_count])
    train_indices.extend(indices[val_count:])
rng.shuffle(train_indices)
rng.shuffle(val_indices)
return train_indices, val_indices
def _build_train_sampler(dataset: datasets.ImageFolder, train_indices: List[int]) ->
WeightedRandomSampler:
class_counts: Dict[int, int] = defaultdict(int)
for idx in train_indices:
    class_counts[dataset.targets[idx]] += 1
sample_weights = [1.0 / class_counts[dataset.targets[idx]] for idx in train_indices]
```

```
    return WeightedRandomSampler(
        weights=torch.DoubleTensor(sample_weights),
        num_samples=len(train_indices),
        replacement=True,
    )
```

train_models.py

```
import argparse
import random
import sys
from pathlib import Path
from typing import Dict
import torch
import torch.nn as nn
import torch.optim as optim
PROJECT_ROOT = Path(__file__).resolve().parent.parent
if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))
from dataset_utils import load_datasets, save_class_indices
from webapp.model_def import RetinalCNN
def set_seed(seed: int) -> None:
    random.seed(seed)
    torch.manual_seed(seed)
```

```
    if torch.cuda.is_available():
        torch.cuda.manual_seed_all(seed)
def evaluate(model, data_loader, criterion, device, num_classes: int):
    model.eval()
    running_loss = 0.0
    correct = 0
    total = 0
    class_correct = [0 for _ in range(num_classes)]
    class_total = [0 for _ in range(num_classes)]
    with torch.no_grad():
        for images, labels in data_loader:
            images = images.to(device)
            labels = labels.to(device)
            outputs = model(images)
            loss = criterion(outputs, labels)
            running_loss += loss.item() * images.size(0)
            _, predicted = torch.max(outputs, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()
            for actual, pred in zip(labels.tolist(), predicted.tolist()):
                class_total[actual] += 1
                if actual == pred:
```

```
                class_correct[actual] += 1
    epoch_loss = running_loss / max(total, 1)
    epoch_acc = correct / max(total, 1)
    per_class_acc = {}
    for class_idx in range(num_classes):
        per_class_acc[class_idx] = class_correct[class_idx] / max(class_total[class_idx],
1)
    balanced_acc = sum(per_class_acc.values()) / max(num_classes, 1)
    return {
        "loss": epoch_loss,
        "accuracy": epoch_acc,
        "balanced_accuracy": balanced_acc,
        "per_class_accuracy": per_class_acc,
    }
def _format_per_class_accuracy(per_class_accuracy: Dict[int, float], idx_to_class:
Dict[int, str]) -> str:
    parts = []
    for idx in sorted(idx_to_class):
        label = idx_to_class[idx]
        score = per_class_accuracy.get(idx, 0.0)
        parts.append(f"{label}={score:.4f}")
    return " | ".join(parts)
```

```
def _compute_class_weights(train_loader, num_classes: int, device: torch.device) ->
torch.Tensor:
  subset = train_loader.dataset
  base_dataset = subset.dataset
  subset_indices = subset.indices
  class_counts = torch.zeros(num_classes, dtype=torch.float32)
  for sample_idx in subset_indices:
      class_idx = base_dataset.targets[sample_idx]
      class_counts[class_idx] += 1.0
  class_counts = torch.clamp(class_counts, min=1.0)
  inv_freq = 1.0 / class_counts
  weights = inv_freq / inv_freq.sum() * num_classes
  return weights.to(device)
def train_single_model(
  dataset_path: Path,
  model_path: Path,
  class_index_path: Path,
  epochs: int = 15,
  batch_size: int = 32,
  learning_rate: float = 3e-4,
  validation_split: float = 0.2,
  seed: int = 42,
```

```
  label_smoothing: float = 0.0,
  ):
  train_loader, val_loader, class_indices = load_datasets(
      dataset_dir=dataset_path,
      batch_size=batch_size,
      validation_split=validation_split,
      seed=seed,
    )
  idx_to_class = {idx: name for name, idx in class_indices.items()}
  print(f"[INFO] Classes detected in {dataset_path}: {sorted(class_indices.keys())}")
  device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
  model = RetinalCNN(num_classes=len(class_indices)).to(device)
  class_weights = _compute_class_weights(train_loader, len(class_indices), device)
    criterion = nn.CrossEntropyLoss(weight=class_weights,
label_smoothing=label_smoothing)
    optimizer = optim.AdamW(model.parameters(), lr=learning_rate,
weight_decay=1e-4)
  scheduler = optim.lr_scheduler.ReduceLROnPlateau(
      optimizer,
      mode="min",
      factor=0.5,
      patience=2,
```

```
    min_lr=1e-6,
  )
best_val_loss = float("inf")
best_val_balanced_acc = 0.0
best_state = None
best_metrics = None
for epoch in range(epochs):
    model.train()
    train_loss = 0.0
    train_correct = 0
    train_total = 0
    for images, labels in train_loader:
        images = images.to(device)
        labels = labels.to(device)
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)
        optimizer.step()
        train_loss += loss.item() * images.size(0)
        _, predicted = torch.max(outputs, 1)
```

```
    train_total += labels.size(0)
    train_correct += (predicted == labels).sum().item()
epoch_train_loss = train_loss / max(train_total, 1)
epoch_train_acc = train_correct / max(train_total, 1)
val_metrics = evaluate(
    model=model,
    data_loader=val_loader,
    criterion=criterion,
    device=device,
    num_classes=len(class_indices),
  )
scheduler.step(val_metrics["loss"])
current_lr = optimizer.param_groups[0]["lr"]
print(
    f"Epoch {epoch + 1}/{epochs} | "
    f"train_loss={epoch_train_loss:.4f} train_acc={epoch_train_acc:.4f} | "
    f"val_loss={val_metrics['loss']:.4f} val_acc={val_metrics['accuracy']:.4f} "
    f"val_bal_acc={val_metrics['balanced_accuracy']:.4f} lr={current_lr:.6f}"
  )
is_better = (
    val_metrics["balanced_accuracy"] > best_val_balanced_acc
    or (
```

```
            val_metrics["balanced_accuracy"] == best_val_balanced_acc
            and val_metrics["loss"] < best_val_loss
        )
    )
def main():
parser = argparse.ArgumentParser(description="Train OCT and Retinal CNN models (PyTorch)")
parser.add_argument("--dataset-root", default="DataSet", help="Path to dataset root folder")
parser.add_argument("--epochs", type=int, default=15, help="Number of training epochs")
parser.add_argument("--batch-size", type=int, default=32, help="Batch size")
parser.add_argument("--lr", type=float, default=3e-4, help="Learning rate")
parser.add_argument("--validation-split", type=float, default=0.2, help="Validation split ratio")
parser.add_argument("--seed", type=int, default=42, help="Random seed")
parser.add_argument("--label-smoothing", type=float, default=0.0, help="Label smoothing for CE loss")
args = parser.parse_args()
set_seed(args.seed)
root = Path(args.dataset_root)
oct_dataset = root / "OCT"
retinal_dataset = root / "RETINAL"
```

```
    if not oct_dataset.exists() or not retinal_dataset.exists():
    raise FileNotFoundError("Expected dataset folders: DataSet/OCT and
DataSet/RETINAL")
    models_dir = Path("models")
    oct_metrics = train_single_model(
        dataset_path=oct_dataset,
        model_path=models_dir / "oct_model.pt",
        class_index_path=models_dir / "oct_class_indices.json",
        epochs=args.epochs,
        batch_size=args.batch_size,
        learning_rate=args.lr,
        validation_split=args.validation_split,
        seed=args.seed,
        label_smoothing=args.label_smoothing,
    )
    )
    print(
        "[SUMMARY] "
        f"OCT val_acc={oct_metrics['accuracy']:.4f}, "
        f"OCT val_bal_acc={oct_metrics['balanced_accuracy']:.4f} | "
        f"RETINAL val_acc={retinal_metrics['accuracy']:.4f}, "
        f"RETINAL val_bal_acc={retinal_metrics['balanced_accuracy']:.4f}"
```

```
    )

if __name__ == "__main__":

    main()
```

app.py

```
from webapp import create_app

app = create_app()

if __name__ == "__main__":

    app.run(host="127.0.0.1", port=5000, debug=False, use_reloader=False)
```

APPENDIX – B

SCREENSHOTS

Sample Output

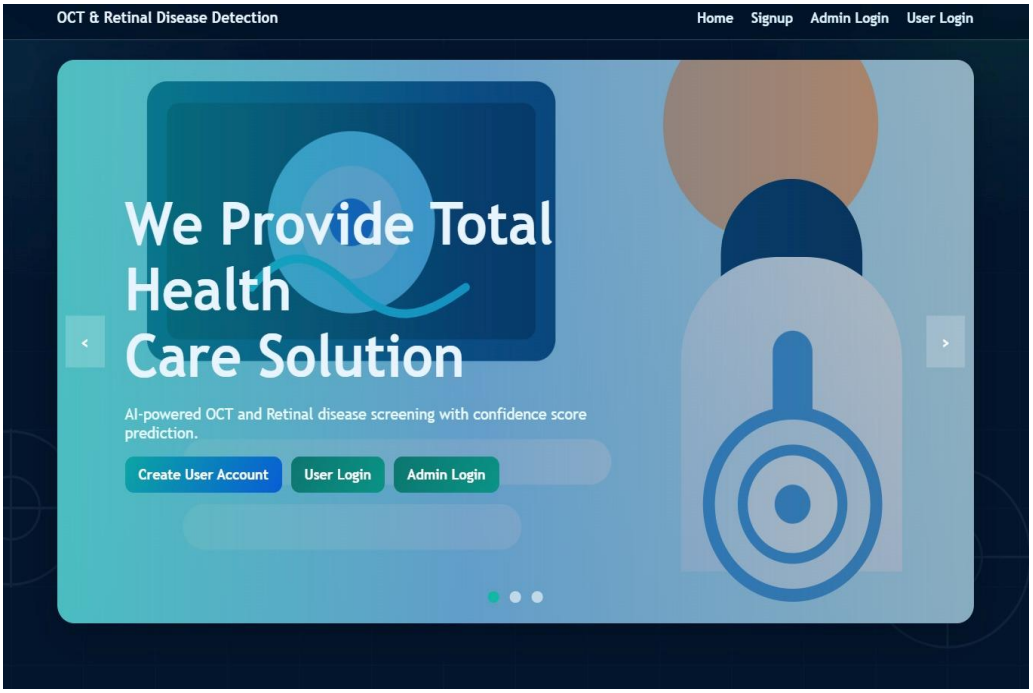

Figure. B.1. Home Page

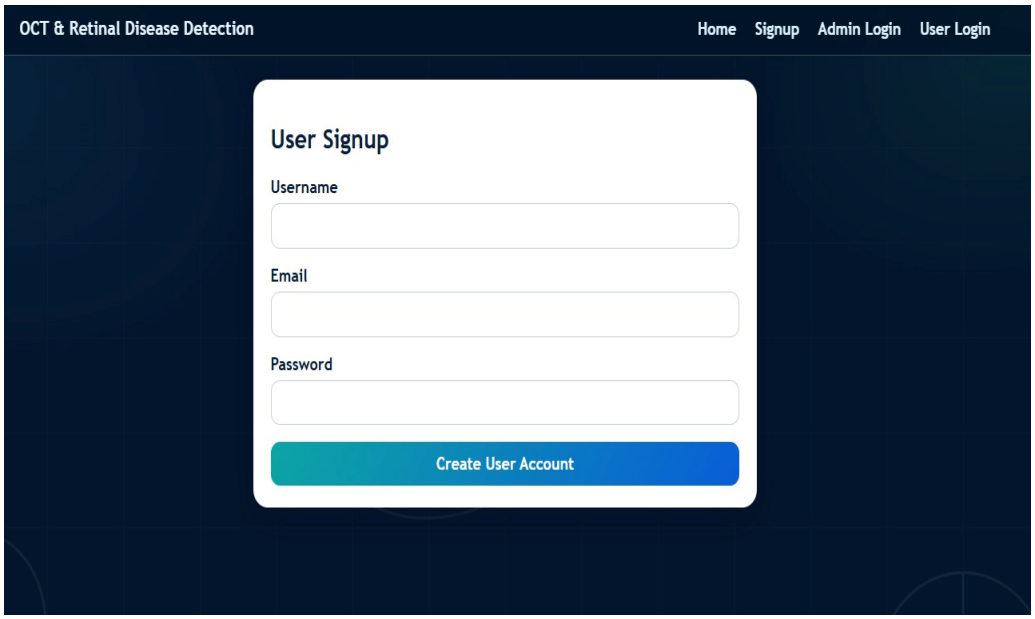

Figure. B.2. Signup Page

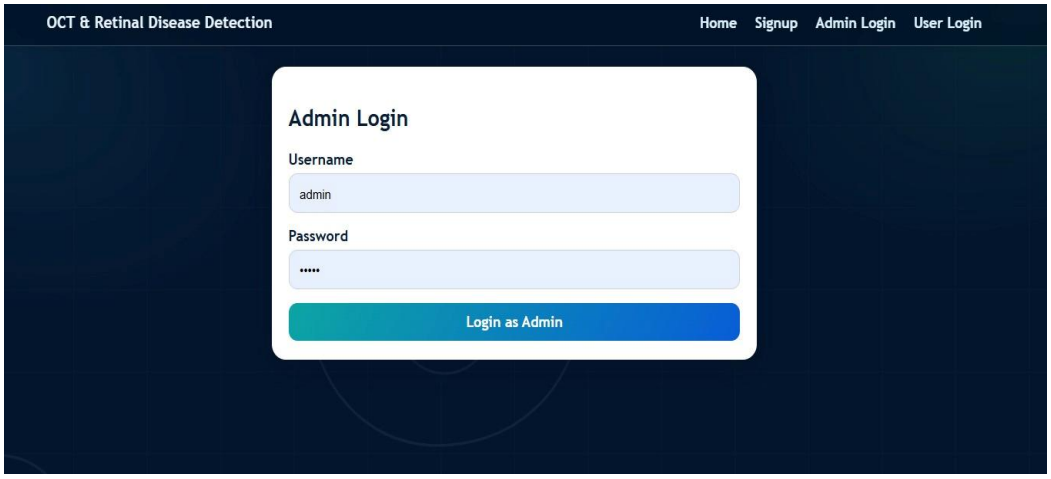

Figure. B.3. Admin Login Page

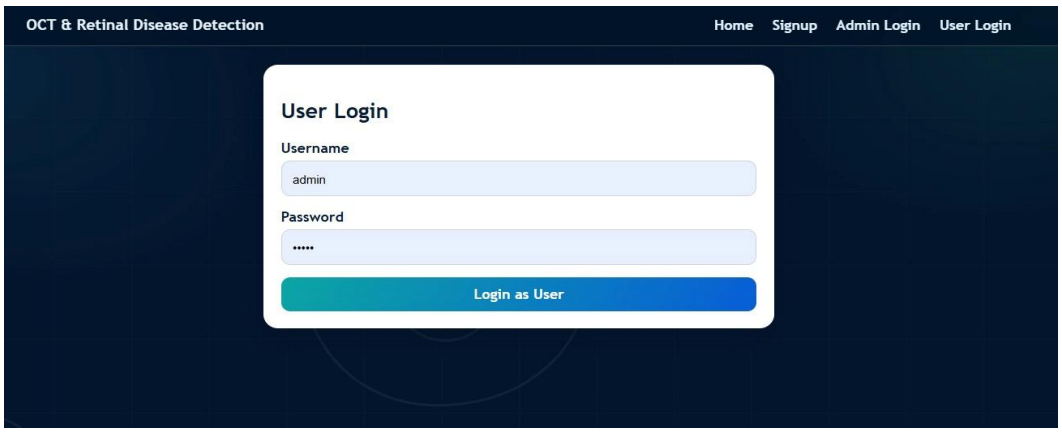

Figure. B.4.User Login Page

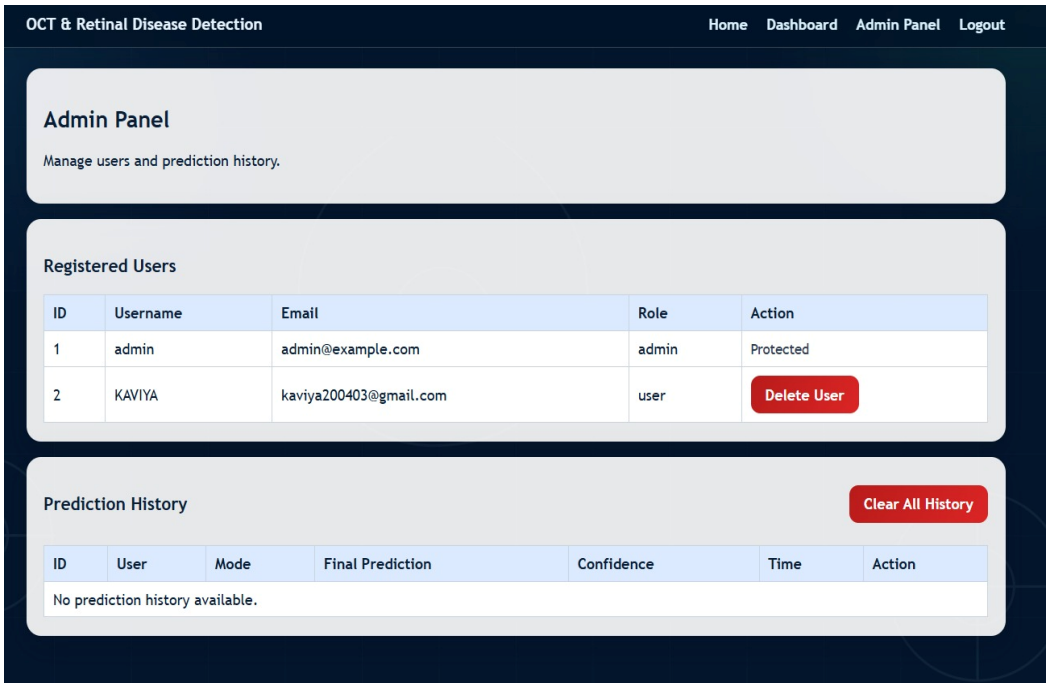

Figure. B.5.Admin Panel

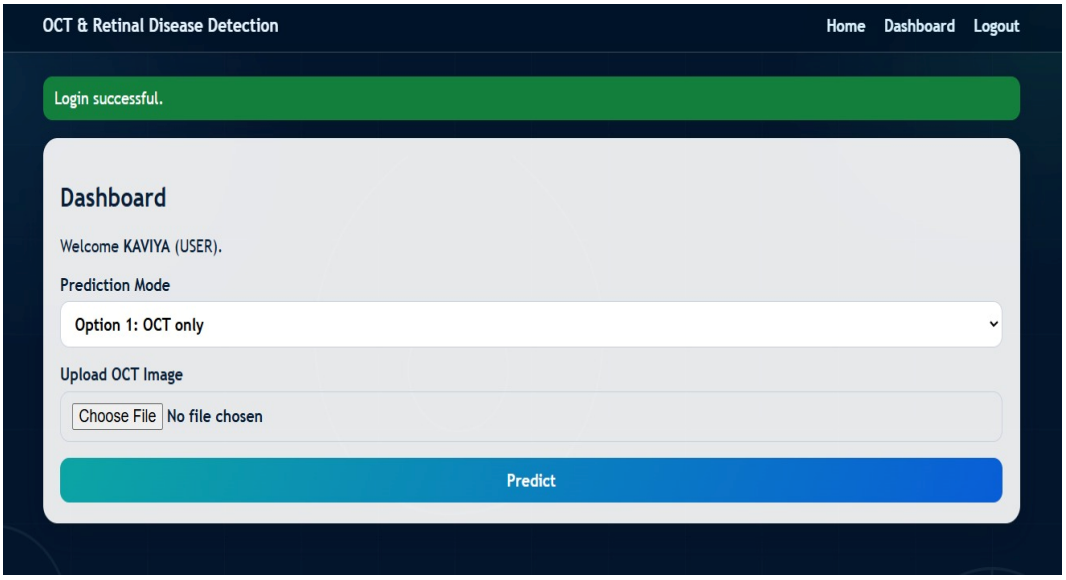

Figure. B.6. User Dashboard

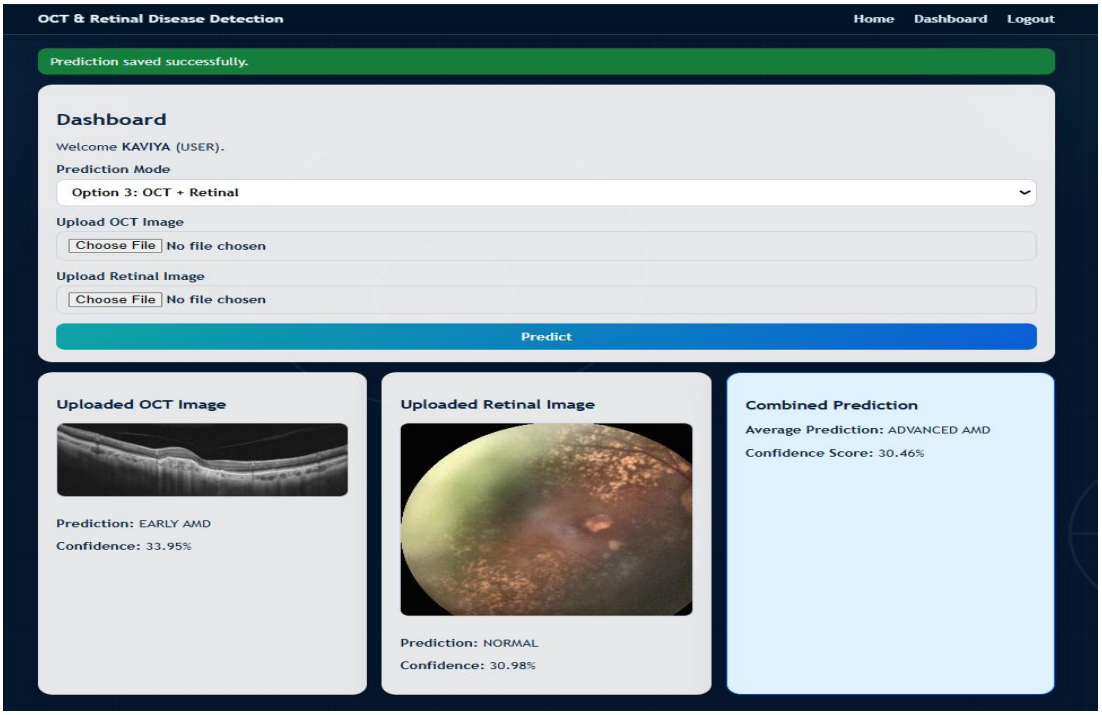

Figure. B.7. Prediction Dashboard

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